



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Experiment No. 7

Title: Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

Software/Tools Required: HTML5, CSS3, JavaScript, a web browser such as Google Chrome or Microsoft Edge, and a code editor such as Visual Studio Code.

Theory

1. DOM Traversal and Selection

The Document Object Model (DOM) represents an HTML document as a tree of elements. JavaScript can traverse and access elements using methods such as `getElementById()`, `querySelector()`, `closest()`, and `querySelectorAll()`. These methods allow a program to find and work with specific elements in a web page.

2. DOM Updates

DOM updates allow JavaScript to dynamically change page content and structure. Elements can be created with `createElement()`, inserted using `appendChild()`, and removed using `remove()`. Properties such as `textContent`, `value`, and `className` can also be modified to update the interface.

3. Adding Tasks

In the to-do list application, the `addTask()` function reads and trims the text entered in the task input. If the input is empty, an alert is displayed. Otherwise, a new list item is created dynamically, containing the task text along with Edit and Delete buttons, and it is appended to the task list.

4. Editing and Deleting Tasks

The `editTask()` function finds the selected list item using `closest('li')` and locates its task text using `querySelector()`. A prompt allows the task to be changed, while empty updates are rejected. The `deleteTask()` function finds the selected list item and removes it from the DOM. When no tasks remain, a “No tasks yet” message is added.

5. Event Handling

JavaScript event handling makes the application interactive. The Add button calls `addTask()`, while Edit and Delete buttons call their respective functions. A keydown event listener also allows the Enter key to add a task. The input is cleared and focused again after a task is added.

Case Study Implemented

The case study is a Login Page form implemented using JavaScript. The script selects the signup form and its Password, Confirm Password, Birthday, Website, and form-status elements. It sets the maximum birthday to the current date, gives website-entry guidance on focus, and checks the website format when its value changes. During submission, it prevents the default action, clears previous messages, verifies that the two passwords match, checks overall form validity, reports invalid fields, and displays either an error message or a success message indicating that the account details are ready to be submitted.

Conclusion

The case study successfully demonstrates DOM traversal, dynamic DOM updates, event handling, and JavaScript validation to create an interactive and user-friendly web application.